Higher National Diploma in Civil Engineering

4th Year, First Semester Examination- 2019

CE3225 –Building and Structures

Instructions for Candidates:
Answer five question only

No. of questions: 06
No. of pages: 03
Time: 03 hours

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01. i) Draw the air conditioning cycle and name the main components. (06 marks)

ii) Name 4 types of air conditioning systems and briefly explain two of them. (06 marks)

iii) Name four possible combinations of natural and mechanical ventilation. (02 marks)

iv) Describe Active and Passive temperature controlling method and give three examples of each method. (06 marks)

02. i) What is the purpose of providing circulation in a multistory building? (02 marks)

ii) List three examples for each horizontal and vertical circulation in a building. (06 marks)

iii) State four types of stair cases and sketch the main component of a staircase. Mention UDA dimension for stairs in a public building. (08 marks)

iv) List four considerations to be required for placing an emergency exit. (04marks)

03. i) List out four reasons of fire in a multi-story building. (04marks)

ii) Classify the types of fire with examples and mention relevant extinguished agents for each fire class. (08 marks)

iii) Briefly describe two types of sprinkler systems. (04marks)

iv) Discuss the importance of structural design to minimize/prevent the fire risk in a building. (02 marks)

v) State all applicable methods to achieve structural strength against fire risk in a building? (02 marks)

04. i) List out three advantages and three disadvantages of sky scrapers. (06 marks)

ii) Briefly describe the function of following sanitary appliances with the aid of sketches.

a) Wash basin.
b) Bath tub.
c) Water closet. (06 marks)

iii) Describe the main two cold water supply system to a multi-story building from the mains. (04 marks)

iv) What are the available methods for protecting a multistory building against wind effect? (04 marks)

05. i) Briefly describe the purpose of "day lighting" and list out 3 number of day lighting methods. (04 marks)

ii) What are the basic aims of safe electrical supply installation? (03 marks)

iii) Discuss the importance of having an "electrical supply and installation drawing" in a multistory building. (03 marks)

iv) Briefly explain the following terms.

a.) Fuse.
b.) Circuit breakers.
c.) Lumen-lm.
d.) Lux-lx
e.) Glare

(10 marks)

06. i) Discuss the advantages of green building over conventional building.

(04 marks)

ii) What are the considerations to be taken in to account in green building construction?

(05 marks)

iii) Briefly describe the term 'Qualified Person' in building regulations with relevant examples.

(03 marks)

iv) Describe the followings with the aid of sketches.

a) Building line.
b) Street line.
c) Turning line.
d) Clearance from electric lines. (08 marks)